AR
2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

IonQ

NYSE : IONQ · Young-Company DCF (Damodaran)

$26.17B mkt cap · 373M sh · $2.39B cash, zero debt · $470M RPO (+554% YoY) at Q1 2026; SkyWater acq. announced Jan '26 (~$1.8B cash+stock; closes Q2/Q3 2026)

29 May 2026 · the qualitative

$70.14

THE CENTRAL QUESTION

Can IonQ capture meaningful share of a $50B global quantum TAM by 2036 with its trapped-ion stack — or is the $26B market cap pricing certainty Quantinuum, Google, and neutral-atom don't support?

IonQ trades at ~$26B against $64.7M Q1 2026 revenue and FY26 guide $260-270M — P/S >95×. $2.39B cash + $470M RPO + the $1.8B SkyWater chip-fab acquisition is the optionality. Trapped-ion gives ~99.9%+ 2Q fidelity; AQ#64 on Tempo hit three months early. But the competitive set is brutal: Quantinuum IPOs at $12.7B with better logical-qubit performance, Google's Willow demonstrated verifiable advantage on chemistry, and neutral-atom scaled past 1,200 qubits. Honest math: cash-feasible raise schedules force heavy dilution before commercial inflection. Five scenarios, weighted; show your work.

PROBABILITY-WEIGHTED EXPECTED VALUE

$9.18 expected fair value today

-86.9% vs spot $70.14

FORWARD COMPOUNDED VALUE

+5y	+10y	+15y	+20y
$13.15	$18.86	$27.07	$38.87
0.19× spot	0.27× spot	0.39× spot	0.55× spot

WEIGHTING RATIONALE

Ultra Bear 10% Trapped-ion scaling wall + Quantinuum dominance + commercial quantum advantage delayed past 2030; equity wipes via dilution.

Bear 30% Quantinuum's IPO + better logical qubits push IonQ to #2 within trapped-ion; smaller TAM than consensus.

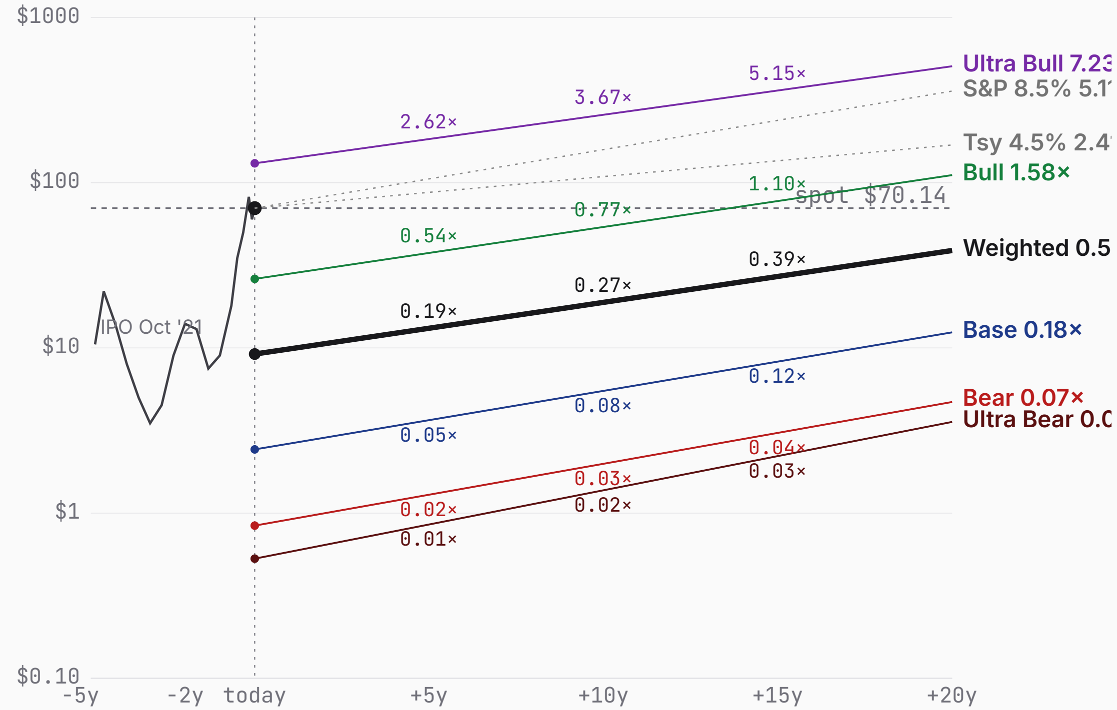
Base 42% Sequential execution on roadmap; co-leader duopoly with Quantinuum at ~8% TAM share and 17% mature margin.

Bull 15% Vertical SkyWater integration pays off; IonQ wins trapped-ion premium and platform race.

Ultra Bull 3% Tail of tails: Nvidia-of-quantum + TAM upside + competitor stumbles + multiple expansion.

Bull (15%) + Ultra Bull (3%) together contribute $7.86 — 86% of the $9.18 expected value despite 18% combined probability. Today's spot ($70.14) sits 664% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED



ULTRA BEAR	BEAR	BASE	BULL	ULTRA BULL
$0.53	$0.84	$2.43	$26.17	$130.99
-99.2% vs spot	-98.8% vs spot	-96.5% vs spot	-62.7% vs spot	+86.8% vs spot
TAM 0.7% P(fail) 60% S2C 0.6 Prob 10%	TAM 2.8% P(fail) 18% S2C 1.0 Prob 30%	TAM 7.6% P(fail) 12% S2C 1.3 Prob 42%	TAM 16.0% P(fail) 8% S2C 1.5 Prob 15%	TAM 32.0% P(fail) 4% S2C 1.7 Prob 3%
Trapped-ion sidelined; equity wiped.	Quantinuum wins; IonQ #2 niche.	Co-leader w/ Quantinuum; ~17% mgn.	Trapped-ion wins; full-stack mgn.	Dominant quantum pure-play; software-premium re-rate.

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IonQ scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 10% Bear 30% Base 42% Bull 15% Ultra Bull 3% · Weighted expected $9.18 (-86.9% vs spot)

ULTRA BEAR $0.53

-99.2% vs spot

Probability 10%

Trapped-ion sidelined; equity wiped.

WHY 10%

10% reflects three compounding tail risks specific to trapped-ion: fundamental scaling ceiling around 100-500 qubits — credible physicists still debate this; Quantinuum's commercial dominance within the trapped-ion category after its $12.7B IPO; the broader risk that "useful" quantum advantage stays elusive through 2030, starving the whole sector. Each individually 25-35%; conjunctive 8-12%. SkyWater acquisition adds $1.8B of fab opex risk if quantum demand doesn't materialize.

WHAT HAPPENS

Structural failure. Quantum advantage on commercial problems takes longer than the bull-deck slides project (BCG's revised "broad advantage 2030-2040" view); the NISQ era stays a science-fair through 2030. IonQ burns $400-600M annually to AQ#256 with no commensurate revenue ramp. Worse, trapped-ion architecture hits a fundamental scaling wall around AQ#100: ion-shuttling complexity, vacuum-chamber engineering, and laser-stability budgets all degrade nonlinearly. Quantinuum's H-series, with the same physics and Honeywell's industrial-scale manufacturing, edges ahead and consolidates the trapped-ion niche.

Meanwhile superconducting (IBM Nighthawk, Google Willow) achieves first verified commercial advantage at a price IonQ can't match per-shot. Neutral-atom scales past 2,000 atoms by FY28 and wins the cost curve. IonQ's $470M RPO converts but never grows; FY30 revenue stalls below $400M against $500M+ cash burn including SkyWater fab opex. Cash exhausts mid-2029. Successive raises in 2028-2030 at $40 → $15 → $4 dilute holders 4×. By FY32 the SkyWater fab is sold for parts; IonQ taken private at a token premium. With 60% within-scenario p_fail, distress recovery ~$0.80. Expected per share ~$0.50.

BEAR $0.84

-98.8% vs spot

Probability 30%

Quantinuum wins; IonQ #2 niche.

WHY 30%

Quantinuum's IPO at $12.7B with materially better logical-qubit performance (48 logical at 99.92%) is a real competitive shift — Honeywell mfg + Cambridge Quantum SW is the most credible full-stack competitor. The broader quantum TAM consistently underperforms forecasts. IonQ executes technically but the pie is smaller and Quantinuum eats the premium slice. Within-scenario 18% p_fail reflects that even in the bear world, IonQ's cash position and DARPA contracts likely prevent bankruptcy. 30% weight reflects high-base-rate competitive entry has already happened.

WHAT HAPPENS

Execution exists but the market shape disappoints. IonQ delivers AQ#256 by 2027 and the SkyWater 200K-qubit chip in 2028 — real milestones. But commercial demand stays government- and research-heavy: DARPA QBI, DOE, NQI, a handful of pharma POCs. Quantinuum's deeper logical-qubit lead wins enterprise mind-share post-IPO; IonQ becomes the credible #2 in trapped-ion, capturing premium government accounts but losing the big AWS/Azure/GCP enterprise-deployment race to superconducting.

Revenue ramps to ~$1.4B FY36 — strong absolute growth but tiny vs bull-deck $5B+. SkyWater fab is a mixed asset: useful internally but underutilized as a third-party foundry, dragging margins. Operating margin matures around 8% — better than legacy services but well below cloud-software economics the current multiple implies. Cash burn averages $300-400M through FY29; two raises ($400M + $1.0B) at $50-30 prices add ~25M shares net. Final ~440M. With 18% within-scenario p_fail, DCF/share ~$5; expected ~$4.20.

BASE $2.43

-96.5% vs spot

Probability 42%

Co-leader w/ Quantinuum; ~17% mgn.

WHY 42%

Highest single bucket because the base requires no breakthrough — sequential execution on plans in motion (Tempo shipping, SkyWater vertical integration, AQ roadmap ahead of schedule, DARPA QBI participation, RPO converting). Coexistence with Quantinuum mirrors duopoly outcomes in adjacent deep-tech markets (AMD/Intel, ASML/Nikon). 42% sits below typical 50-55% base weights because quantum has more genuine tail risk than most categories — thin middle ground between "works commercially" and "doesn't." Five- scenario structure with fatter tails forces base weight down.

WHAT HAPPENS

Modal outcome. IonQ executes the published roadmap roughly on schedule: AQ#64 done, AQ#256 by 2027, SkyWater fab chips back in 2028, the 200K-qubit-chip class reached 2028-29. Quantinuum is a credible co-leader; the trapped-ion sub-segment becomes a duopoly (~50/50 share), with IonQ winning roughly half the US government, defense, and large-enterprise quantum-cloud business via existing AWS Braket / Azure Quantum / GCP integration. Neutral-atom takes cost-sensitive optimization; superconducting takes most "deep stack" enterprise via IBM/Google brand.

By FY36 IonQ does ~$3.8B revenue (~7.6% of $50B TAM) at 17% mature operating margin — better than legacy hardware but below cloud-SaaS. SkyWater fab supplies internal demand plus third-party defense- electronics revenue. RPO converts steadily; multi-product customer share rises 35% → 60%+. Cash burn troughs at ~$350M in FY28; a $2.5B raise programme across years 1-7 at $35-75 keeps cash≥0. SkyWater stock-portion dilution (~25M) folded in. Final shares ~450M. DCF/share ~$16; with 12% within-scenario p_fail, expected ~$14.70 — well below today's $70 spot.

BULL $26.17

-62.7% vs spot

Probability 15%

Trapped-ion wins; full-stack mgn.

WHY 15%

Chain of conditionals: AQ#256 ~on schedule (~70%), SkyWater 200K-qubit chip delivered 2028 (~60%), IonQ — not Quantinuum — captures the trapped-ion premium position (~50%), trapped-ion beats neutral-atom and superconducting on commercially relevant workloads at scale (~50%), DARPA/DOE flows weighted to IonQ (~60%). Each individually plausible; joint ~6-8% before correlation; correlated upward to ~12-18% because conditional outcomes reinforce. 15% is a real but non-modal tail.

WHAT HAPPENS

Trapped-ion is correct architecture and SkyWater compounds. By 2028 vertical integration via SkyWater proves out: the 200K-qubit chip arrives on schedule, error correction works, and IonQ demonstrates verifiable commercial quantum advantage on a customer workload (likely chemistry or optimization for a Fortune 50). Quantinuum stays credible #2 within trapped-ion but IonQ's vertical fab + cloud-native distribution wins the platform race. Neutral-atom struggles with fault tolerance; superconducting stalls between AQ-equivalent #500 and #2000 due to error-correction overhead.

Revenue ramps to ~$8B FY36 (~16% TAM share) with 27% mature operating margin — the SkyWater fab is now monetizing third-party customers as the dominant US trusted quantum-chip foundry. DARPA HARQ scales into a multi-billion national-security program. Federal NQI funding flows disproportionately to IonQ. Cash burn troughs FY28 then turns positive by FY30. A $3.4B raise programme across years 1-7 at $50-90 keeps cash≥0. Final shares ~450M. DCF/share ~$54; with 8% within-scenario p_fail, expected ~$50.

ULTRA BULL $130.99

+86.8% vs spot

Probability 3%

Dominant quantum pure-play; software-premium re-rate.

WHY 3%

Conjunctive bull chain × TAM upside × multiple expansion × competitor collapse. Each piece individually 40-60%; joint ~3-5%. Genuine tail capturing the world where quantum becomes a category- defining tech platform and IonQ is the dominant US pure-play. The asymmetrical-moonshots tier's reason-for-being: real upside that you can't get to without it.

WHAT HAPPENS

Tail of tails. Every Bull condition hits AND the FY36 TAM is materially larger than consensus (Roots Analysis $198B FY40 trajectory, McKinsey upper-bound $72B by 2035 just for computing) AND IonQ achieves de facto platform monopoly in trapped-ion + becomes a meaningful chip foundry via SkyWater. Quantinuum struggles with IPO-era execution; superconducting hits fault- tolerance walls that trapped-ion's higher native fidelity sidesteps. IonQ becomes the "Nvidia of quantum" — vertically integrated, full-stack, with AWS/Azure/GCP installed customer base creating a software flywheel.

By FY36 IonQ does $16B+ revenue at 33% mature operating margin (full-stack hardware + cloud + software-take-rate on partner apps). SkyWater fab serves the entire US national-security stack. Multiple expansion to software-hybrid premium. Modest $3.1B raise in years 1-6 at $70-100 maintains runway. Final shares ~435M. DCF/share ~$135; with 4% p_fail, expected $130.

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

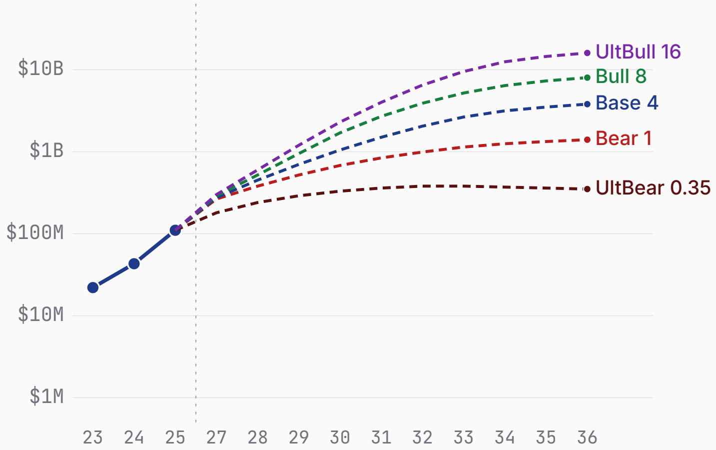
29 May 2026 · the snapshot

IonQ business snapshot

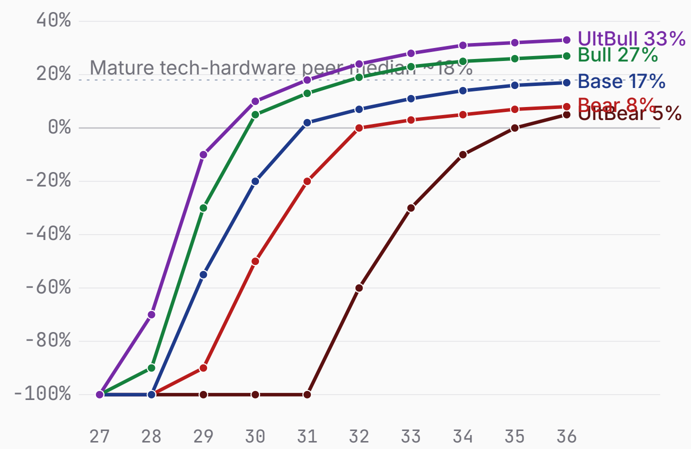
Bear $0.84 Base $2.43 Bull $26.17

FY23-FY25 history + FY26 (guide) + FY27-FY36 scenario projections · fiscal years end Dec 31 · 10-K FY25, Q1 2026 10-Q, McKinsey Quantum Monitor 2026

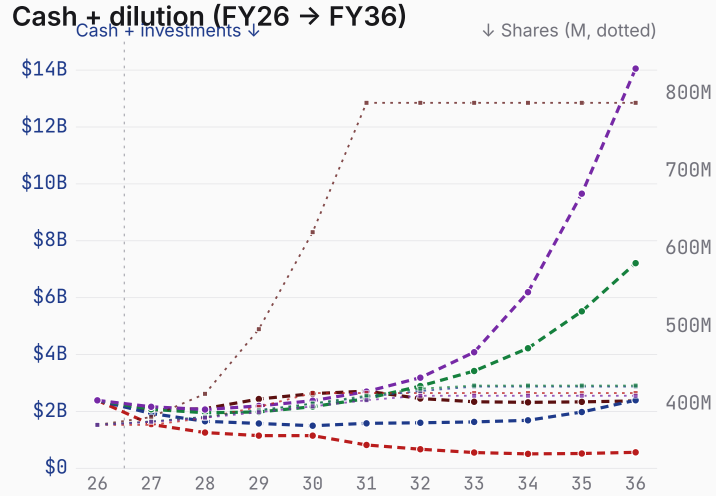
Revenue — history + scenarios to FY36 (log)



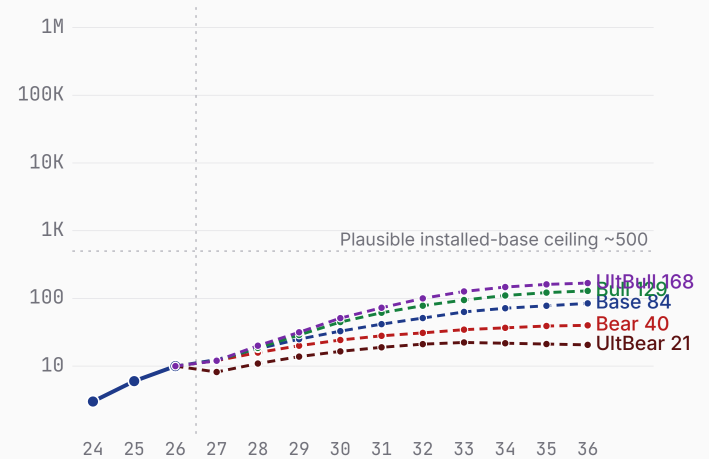
Path to profitability — op margin (FY27→FY36)



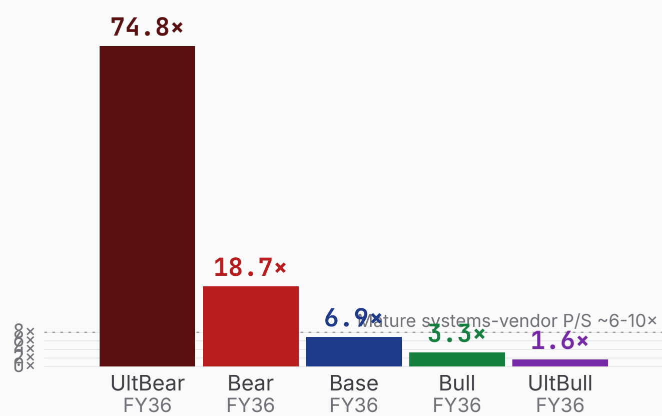
Cash + dilution (FY26 → FY36)



AQ#20+ class systems deployed (log)

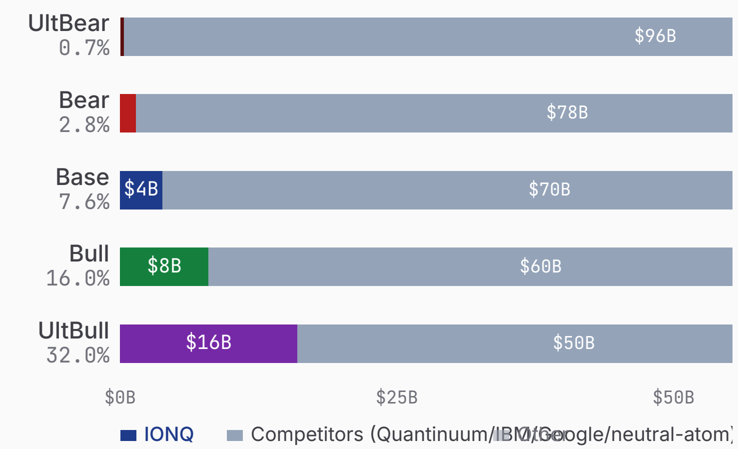


$26.17B mkt cap as P/S on FY36 rev



Bear=fantasy, base=stretched, bull=full, ultra=cheap on FY36 rev.

$50B global quantum TAM (FY36) — IONQ share



Sources: IONQ Q1 2026 10-Q, FY25 10-K, IonQ Q1'26 earnings (8-K), Quantinuum S-1, McKinsey Quantum Technology Monitor 2026, BCG quantum value-creation revision, DARPA QBI Stage B selections (Nov 2025).

IonQ 7 Powers · the moat, scored

Bear $0.84 Base $2.43 Bull $26.17

Arena. The race to a ~$50B FY36 fault-tolerant quantum-computing TAM — trapped-ion (IonQ, Quantinuum) vs. superconducting (Google, IBM) vs. neutral-atom (QuEra, Pasqal, Atom Computing).

POWER ORIGINATION
origination window: open

POWER ORIGINATION · 7 POWERS

Scale Economies	1/3	SkyWater fab vertical integration begun; not yet a cost-curve advantage.
Network Economies	0/3	No developer/data flywheel; cloud reach via AWS/Azure is rented, not owned.
Counter-Positioning	2/3	Trapped-ion stack incumbents can't pivot to without abandoning sunk superconducting fabs.
Switching Costs	2/3	$470M RPO; algorithms compiled to IonQ's native gate set raise re-port cost.
Branding	1/3	Strong research credibility (AQ#64 early); not yet the enterprise default.
Cornered Resource · thesis leans here	2/3	Trapped-ion IP portfolio + SkyWater fab; Maryland/Duke talent lineage.
Process Power	1/3	Fidelity yield improving each generation; not yet a durable, opaque process edge.

THE RACE · NAMED RIVALS

Quantinuum — $1.4B+ raised
~20% terminal share · Helios: 48 logical qubits @ 99.92%; $12.7B IPO; Honeywell-backed.

Google Quantum AI — Alphabet balance sheet
~16% terminal share · Willow: verifiable quantum advantage demonstrated.

IBM Quantum — IBM balance sheet
~14% terminal share · Nighthawk roadmap to 1,080 qubits; deep enterprise channel.

Neutral-atom (QuEra/Pasqal/Atom) — $300M+ aggregate
~10% terminal share · 1,200+ physical qubits in 18 months; different error profile.

LEAD / LAG · FALSIFIERS

Algorithmic qubits (AQ)	64 vs 56	LEADING
Logical qubits @ >=99.9%	36 vs 48	LAGGING
Cash runway (yrs at burn)	4 vs 6	LAGGING

> **COMPETITIVE READ**
> Real trapped-ion cornered-resource and counter-positioning leads, but network and scale power are unoriginated and Quantinuum's logical-qubit edge is the falsifier to watch; the origination window is open, but not indefinitely.

IonQ show your work

Bear $0.84 Base $2.43 Bull $26.17 · Weighted $9.18 (-86.9%)

ULTRA BEAR $0.53

ASSUMPTIONS

TAM Year-10 ($B)	50
Mature share (%)	0.7
Mature op margin (%)	5
Sales-to-capital	0.6
Terminal WACC (%)	10.0
Terminal growth (%)	1.5
P(failure) (%)	60
Scenario probability (%)	10

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.18	-300%	-0.64	-0.56
FY28	0.24	-250%	-0.70	-0.54
FY29	0.29	-200%	-0.66	-0.45
FY30	0.33	-150%	-0.56	-0.34
FY31	0.36	-100%	-0.41	-0.22
FY32	0.38	-60%	-0.26	-0.12
FY33	0.38	-30%	-0.11	-0.05
FY34	0.37	-10%	-0.04	-0.01
FY35	0.36	+0%	+0.00	+0.00
FY36	0.35	+5%	+0.02	+0.01

Σ PV explicit FCF	-2.28
+ PV terminal (g 1.5%)	0.07

EQUITY BUILD · $B & PER SHARE

Operating EV	$-2.21B
+ Cash & investments	$2.39B
= Equity value	**$0.18B**
Raised over period	$3.50B
Dilution by FY36	+75%
FY36 shares (M)	1,492
DCF per share	**$0.12**
× (1−P_fail) [40%]	$0.05
+ P_fail × distress [60% × $0.80]	$0.48
= Expected per share	**$0.53**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$0.85	$1.37	$2.21	$3.57
0.01×	0.02×	0.03×	0.05×

BEAR $0.84

ASSUMPTIONS

TAM Year-10 ($B)	50
Mature share (%)	2.8
Mature op margin (%)	8
Sales-to-capital	1.0
Terminal WACC (%)	9.0
Terminal growth (%)	2.0
P(failure) (%)	18
Scenario probability (%)	30

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.27	-220%	-0.70	-0.61
FY28	0.38	-150%	-0.73	-0.57
FY29	0.52	-90%	-0.63	-0.43
FY30	0.68	-50%	-0.49	-0.30
FY31	0.84	-20%	-0.32	-0.18
FY32	0.99	+0%	-0.15	-0.07
FY33	1.14	+3%	-0.08	-0.03
FY34	1.25	+5%	-0.02	-0.01
FY35	1.33	+7%	+0.01	+0.01
FY36	1.40	+8%	+0.04	+0.01

Σ PV explicit FCF	-2.20
+ PV terminal (g 2.0%)	0.21

EQUITY BUILD · $B & PER SHARE

Operating EV	$-1.99B
+ Cash & investments	$2.39B
= Equity value	**$0.40B**
Raised over period	$1.40B
Dilution by FY36	+18%
FY36 shares (M)	440
DCF per share	**$0.91**
× (1−P_fail) [82%]	$0.75
+ P_fail × distress [18% × $0.50]	$0.09
= Expected per share	**$0.84**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$1.29	$1.99	$3.06	$4.71
0.02×	0.03×	0.04×	0.07×

BASE $2.43

ASSUMPTIONS

TAM Year-10 ($B)	50
Mature share (%)	7.6
Mature op margin (%)	17
Sales-to-capital	1.3
Terminal WACC (%)	8.5
Terminal growth (%)	2.5
P(failure) (%)	12
Scenario probability (%)	42

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.28	-200%	-0.69	-0.61
FY28	0.45	-120%	-0.71	-0.56
FY29	0.70	-55%	-0.64	-0.45
FY30	1.05	-20%	-0.56	-0.35
FY31	1.50	+2%	-0.42	-0.24
FY32	2.05	+7%	-0.41	-0.21
FY33	2.65	+11%	-0.31	-0.14
FY34	3.15	+14%	-0.06	-0.03
FY35	3.50	+16%	+0.21	+0.08
FY36	3.80	+17%	+0.35	+0.13

Σ PV explicit FCF	-2.38
+ PV terminal (g 2.5%)	2.13

EQUITY BUILD · $B & PER SHARE

Operating EV	$-0.25B
+ Cash & investments	$1.40B
= Equity value	**$1.15B**
Raised over period	$2.50B
Dilution by FY36	+20%
FY36 shares (M)	450
DCF per share	**$2.56**
× (1−P_fail) [88%]	$2.25
+ P_fail × distress [12% × $1.50]	$0.18
= Expected per share	**$2.43**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$3.65	$5.49	$8.26	$12.42
0.05×	0.08×	0.12×	0.18×

BULL $26.17

ASSUMPTIONS

TAM Year-10 ($B)	50
Mature share (%)	16.0
Mature op margin (%)	27
Sales-to-capital	1.5
Terminal WACC (%)	7.5
Terminal growth (%)	3.0
P(failure) (%)	8
Scenario probability (%)	15

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.29	-180%	-0.68	-0.60
FY28	0.52	-90%	-0.77	-0.61
FY29	0.95	-30%	-0.79	-0.57
FY30	1.70	+5%	-0.75	-0.49
FY31	2.70	+13%	-0.72	-0.42
FY32	3.90	+19%	-0.56	-0.30
FY33	5.20	+23%	-0.10	-0.05
FY34	6.40	+25%	+0.40	+0.18
FY35	7.30	+26%	+1.00	+0.42
FY36	8.00	+27%	+1.46	+0.58

Σ PV explicit FCF	-1.87
+ PV terminal (g 3.0%)	13.19

EQUITY BUILD · $B & PER SHARE

Operating EV	$11.32B
+ Cash & investments	$1.40B
= Equity value	**$12.72B**
Raised over period	$3.40B
Dilution by FY36	+21%
FY36 shares (M)	450
DCF per share	**$28.27**
× (1−P_fail) [92%]	$26.01
+ P_fail × distress [8% × $2.00]	$0.16
= Expected per share	**$26.17**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$37.57	$53.94	$77.43	$111.17
0.54×	0.77×	1.10×	1.58×

ULTRA BULL $130.99

ASSUMPTIONS

TAM Year-10 ($B)	50
Mature share (%)	32.0
Mature op margin (%)	33
Sales-to-capital	1.7
Terminal WACC (%)	7.0
Terminal growth (%)	3.5
P(failure) (%)	4
Scenario probability (%)	3

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.30	-150%	-0.63	-0.56
FY28	0.60	-70%	-0.77	-0.63
FY29	1.20	-10%	-0.83	-0.61
FY30	2.30	+10%	-0.89	-0.60
FY31	4.00	+18%	-0.75	-0.47
FY32	6.50	+24%	-0.20	-0.12
FY33	9.50	+28%	+0.90	+0.47
FY34	12.50	+31%	+2.11	+1.04
FY35	14.50	+32%	+3.46	+1.59
FY36	16.00	+33%	+4.40	+1.89

Σ PV explicit FCF	+2.01
+ PV terminal (g 3.5%)	55.89

EQUITY BUILD · $B & PER SHARE

Operating EV	$57.90B
+ Cash & investments	$1.40B
= Equity value	**$59.30B**
Raised over period	$3.10B
Dilution by FY36	+17%
FY36 shares (M)	435
DCF per share	**$136.32**
× (1−P_fail) [96%]	$130.87
+ P_fail × distress [4% × $3.00]	$0.12
= Expected per share	**$130.99**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$183.72	$257.68	$361.41	$506.89
2.62×	3.67×	5.15×	7.23×

IonQ supporting analysis · disclaimers · glossary

Bear $0.84 Base $2.43 Bull $26.17

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

1 **Best-capitalized pure-play.**
$2.4B cash + $470M RPO + SkyWater vertical integration. Multi-year runway with optionality no other public quantum has.

2 **AQ#64 hit early.**
Tempo achieved AQ#64 three months ahead of schedule — first to do so. Real execution beats slideware.

3 **Federal tailwind is real.**
DARPA QBI Stage B + HARQ + DOE/NQI's $2.7B/5yr authorization disproportionately benefits US-based trusted-supply players.

4 **Trapped-ion fidelity moat.**
Industry-leading 2Q gate fidelity (~99.9%+) is a physics advantage, not process — hard for superconducting/neutral-atom to close.

5 **Cloud distribution unique.**
Available on AWS Braket, Azure Quantum, Google Cloud. Only Quantinuum has comparable reach; IBM/Google don't partner cross-platform.

FALSIFICATION TRIGGERS

Bear validation

AQ#256 slips past 2027 · RPO conversion stalls · SkyWater close delays past Q3 2026 · Quantinuum wins a marquee enterprise displacement

Bull validation

AQ#256 hit by H1 2027 · First verified commercial quantum advantage on customer workload · Multi-product share >50% · 2nd large govt contract (>$100M)

Reframe needed

If neutral-atom (QuEra/Atom Computing) achieves fault-tolerance first — trapped-ion premium thesis erodes; revisit TAM share assumptions across all scenarios

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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Trapped-ion qubits — IonQ's qubit modality — ytterbium ions in electromagnetic traps, manipulated by lasers. Highest gate fidelity but slower clock than superconducting.

AQ (Algorithmic Qubits) — IonQ's benchmark for usable capacity. AQ#N = circuits requiring 2^N states. AQ#64 hit 2026; AQ#256 targeted 2027.

Quantum advantage — Quantum solving a problem faster/cheaper than classical. Google's Willow demonstrated verifiable advantage on chemistry Oct 2025; no commercial advantage verified yet.

Quantinuum — Honeywell-backed full-stack trapped-ion competitor. Filed for $12.7B IPO May 2026; better logical-qubit performance (48 logical at 99.92%).

SkyWater — US semiconductor foundry IonQ is acquiring (~$1.8B, closes Q2/Q3 2026). Domestic chip fab — enables 200K-qubit chip in 2028.

Tempo / Forte — IonQ system generations. Forte = current commercial (AQ#36). Tempo = next-gen, AQ#64 demonstrated, 256-qubit chip.